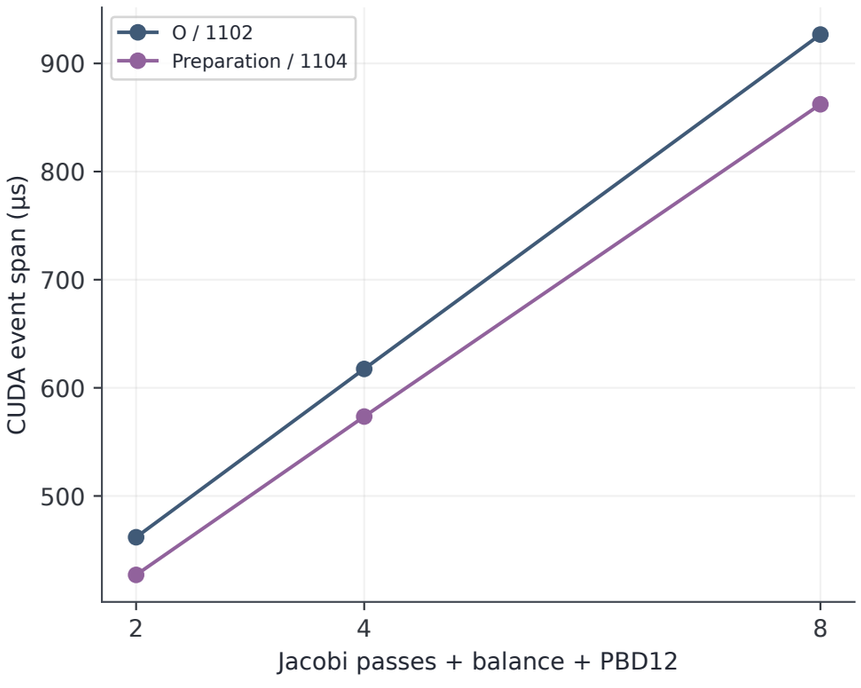
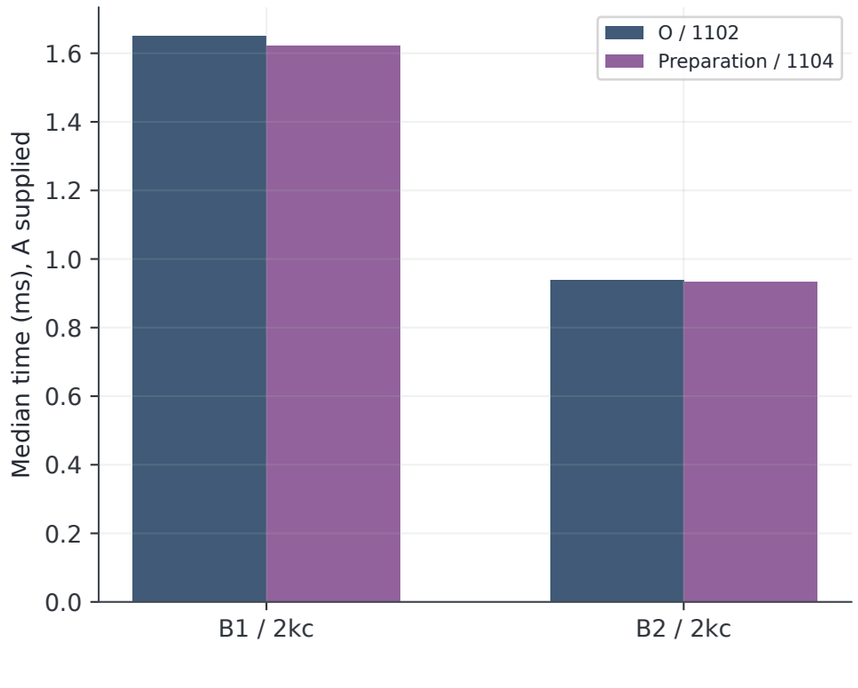
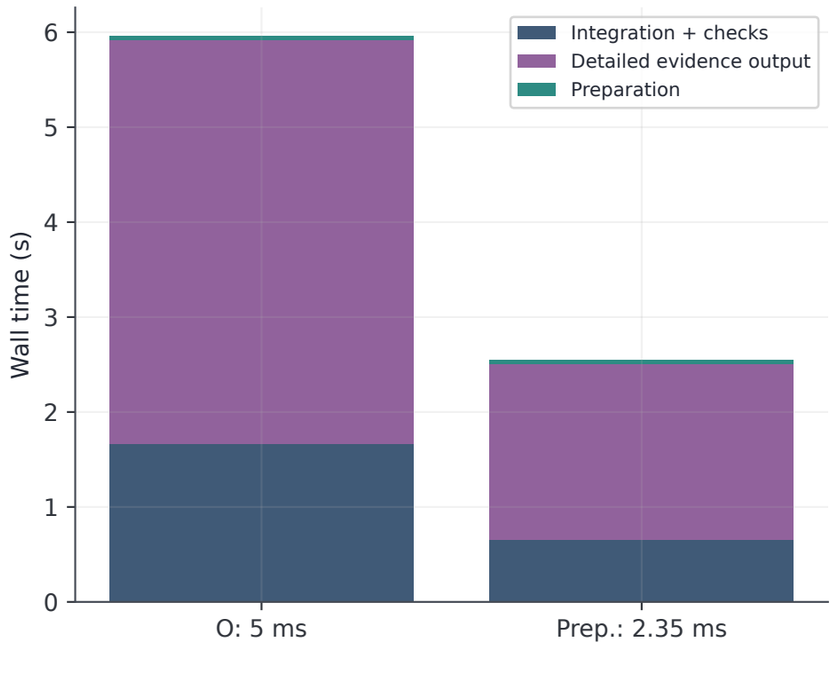
a Original GPU kernels, fixed states



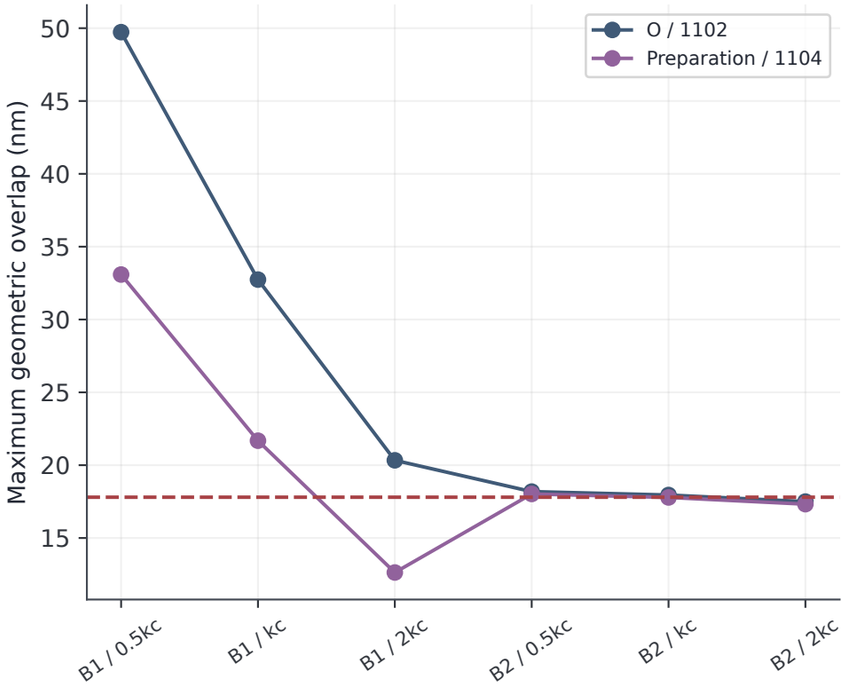
b CPU implicit initialization + contact



c Short-window diagnostic cost



d Two fixed-state contact budgets



e Admission evidence

Check	Status
Discrete lubrication reference	PASS
Two / four bubble contact	PASS
B2 / 2kc: both 32-state checks	PASS
B2 / 2kc: two short windows	PASS
Same-model dt convergence	NOT RUN
Production GPU soft-contact cost	NOT RUN
Exact geometric spring law	NO
Calibrated shell-contact law	NO
Production / AI admission	NO

f Scope and next admission gate

Two short windows completed
735 accepted steps; no contact rejection

Old / new morphology differs:
max. fixed-ID difference 3.657 / 2.515 µm
centroid difference only 1.566 / 0.259 nm

No new 10 ms or OFF/S/O triplet
No production default change
No claim of experimental validation

Next: contact-normal / law refinement,
same-model dt convergence,
then bounded matrix-free GPU feasibility.

Fixed-state kernel spans and CPU reference costs have different scopes: neither is a full production trajectory benchmark.
Detailed observer output is charged. All candidates remain EXPLORATORY / PHYSICS_UNCALIBRATED; training_ready = false.